

Use-Case Flow Specification

Use Case Name: Reserve 30-Minute Slot (UC-02)

Primary Actor: EV Driver

Secondary Actor: Station Operator

Preconditions

1. The EV Driver is logged into the application.
2. The EV Driver has selected a compatible charging socket (CCS, CHAdemo, or Type 2).

Postconditions

1. The selected 30-minute time slot is locked exclusively for the driver.
2. Payment is confirmed, and a digital booking receipt is generated.
3. Socket status updates to "Reserved" across the network.

Main Success Scenario

1. **Select Slot:** EV Driver picks an open 30-minute time slot for a compatible charger.
2. **Calculate Rate:** System calculates dynamic power rates based on current grid demand.
3. **Confirm & Pay:** EV Driver submits payment details to reserve the socket.
4. **Lock Socket:** System authorizes payment and locks the selected socket for the 30-minute window.
5. **Issue Receipt:** System displays confirmation, GPS navigation, and arrival timer.
6. **Plug In:** Driver arrives and plugs in within 10 minutes of slot start time.

Key Alternate Flows

- **Payment Failed:** If payment is declined, the system prompts for an alternate payment method; if not completed within 3 minutes, the slot hold is released.
- **Booking Overlap:** If two drivers request the same slot concurrently, the system assigns it to the first processed request and notifies the second driver to pick another slot.
- **No-Show / Penalty:** If the driver fails to plug in within 10 minutes of the start time, the system releases the socket lock and applies a no-show penalty fee.